

Q-NarwhalKnight

Node Operator Guide

Zero-Config Setup, Fee Structure,
Admin Wallet & OAuth2 Wallet Linking

v2.0

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Abstract

Q-NarwhalKnight (QNK) is a quantum-resistant blockchain with a DAG-Knight consensus engine. Node operators who run public-facing QNK nodes earn **0.1% of every block reward** credited directly to their configured admin wallet, plus a share of DEX protocol fees. As of v8.6.5, new operators can spin up a fully-configured node *in under 60 seconds* using the built-in **Setup Wizard**—no manual environment variables, no hand-written systemd files. This guide covers the zero-config setup wizard, hardware requirements, installation, admin wallet configuration, the 2% development fee structure, OAuth2 device-login wallet linking for miners, the admin API, and the TUI monitoring dashboard.

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1 Introduction

The QNK network relies on a distributed set of bootstrap and relay nodes that store the full blockchain, serve API requests to wallets and miners, relay blocks via libp2p gossipsub, and facilitate turbo-sync for new peers.

Running a node is voluntary, but the protocol now *rewards* operators with a share of the development fee embedded in every coinbase transaction. This creates a sustainable incentive for decentralised infrastructure without requiring staking locks or delegation mechanics.

1.1 Why Run a Node?

- **Earn passive income** – 0.1% of every block reward is credited to your wallet.
- **Earn DEX fees** – 0.05% of every swap on the built-in DEX is shared with operators.
- **Strengthen the network** – more nodes means faster sync, better geographic coverage, and higher censorship resistance.
- **First-class citizen** – operators get the admin API, real-time TUI dashboard, and direct P2P connectivity.

2 System Requirements

Resource	Minimum	Recommended
RAM	4 GB	8 GB+
Disk	50 GB SSD	100 GB+ NVMe
CPU	2 cores x86_64	4+ cores
Network	10 Mbps stable	100 Mbps
OS	Linux x86_64	Ubuntu 22.04+ / Debian 12+

Table 1: Hardware requirements for a QNK node.

2.1 Network Ports

Port	Protocol	Purpose
8080	TCP/HTTP	REST API, SSE streaming, miner submissions, wallet queries
9001	TCP/QUIC	libp2p gossipsub P2P – block relay, peer discovery (Kademlia DHT)

Table 2: Required network ports (open in firewall).

3 Installation & First Boot

3.1 Download the Binary

Download and start (one command)

```

1 # Download the latest release
2 wget https://quillon.xyz/downloads/q-api-server-linux-x86_64
3 chmod +x q-api-server-linux-x86_64
4
```

```

5 # Just run it -- the Setup Wizard handles the rest
6 ./q-api-server-linux-x86_64 --port 8080

```

On first boot (no `.env` file detected), the **Setup Wizard** launches automatically. It takes roughly 60 seconds and requires no terminal expertise. See Section 4 for the full flow.

If you prefer manual configuration, pass `-admin-wallet`:

Manual start (skips wizard)

```

1 ./q-api-server-linux-x86_64 --port 8080 \
2   --admin-wallet <YOUR_64_CHAR_HEX_WALLET>

```

After setup completes, the node will:

1. Load configuration from the generated `.env` file.
2. Create the data directory (`data-mainnet-genesis/` by default).
3. Generate a unique libp2p identity key.
4. Connect to bootstrap peers via Kademlia DHT.
5. Begin syncing the blockchain (turbo-sync for fast catch-up).

3.2 Environment Variables

The Setup Wizard writes these automatically, but you can override any value by editing `.env` or setting shell environment variables.

Variable	Default	Description
Q_ADMIN_WALLET	founder wallet	64-char hex address receiving operator fees
Q_NETWORK_ID	mainnet-genesis	Network identifier (prevents cross-chain contamination)
Q_DB_PATH	./data-{network}	Database storage directory
Q_NODE_OPERATOR_FEE_PROMILLE	50	Operator share in per-mille of dev fee (50 = 5%)
ROCKSDB_BLOCK_CACHE_MB	auto	RocksDB block cache size (auto-tuned by hardware tier)
Q_PREFLIGHT_CHECK	1	Run database integrity check on startup
Q_TURBO_SYNC	1	Enable batched P2P blockchain sync
Q_BOOTSTRAP_URL	https://quillon.xyz	Bootstrap server for setup wizard config fetch

Table 3: Key environment variables (auto-configured by Setup Wizard).

4 Setup Wizard (Zero-Config First Boot)

Starting with v8.6.5, the QNK node ships with an integrated **Setup Wizard** that eliminates manual configuration entirely. New operators download the binary, run it, approve a browser login, and the node is ready.

4.1 When Does It Run?

The wizard triggers automatically when *all three* conditions are met:

1. No `.env` file exists in the working directory.

2. No `-admin-wallet` CLI flag was provided.
3. No `Q_ADMIN_WALLET` environment variable is set.

You can also invoke it manually at any time:

```
1 ./q-api-server --setup
```

4.2 What It Does (4 Steps)



Figure 1: Setup Wizard pipeline.

1. **Detect Hardware** — Reads RAM, CPU cores, and disk type (SSD vs HDD via `/sys/block/*/queue/rot`). Maps to a hardware tier:

Tier	RAM Range	Cache (MB)
Low	≤ 4 GB	512
Medium	4–8 GB	1024
High	8–32 GB	2048
XLarge	> 32 GB	4096

Table 4: Hardware tier mapping.

If an HDD is detected, `Q_CHEAP_SSD=1` is automatically enabled to reduce write amplification.

2. **Fetch Bootstrap Config** — Contacts the bootstrap server at `GET https://quillon.xyz/api/v1/node-config` to retrieve the current network ID, bootstrap peer addresses, and recommended environment variables. If the bootstrap is unreachable, sensible defaults are used.
3. **Browser Login (OAuth2 Device Flow)** — Opens a browser tab at `https://quillon.xyz/miner-login?code=XXXX` where you log in with your wallet. The wizard polls the server every 3 seconds (up to 10 minutes) until your wallet address is confirmed. If login is skipped, the node starts with the founder wallet and you can set `Q_ADMIN_WALLET` in `.env` later.
4. **Write Configuration** — Generates two files:
 - `.env` — all environment variables, hardware-tuned.
 - `q-api-server.service` — a systemd unit file. If running as root, it is installed to `/etc/systemd/system/` and auto-enabled. If not root, it is saved locally with manual install instructions.

4.3 Re-running Setup

The wizard is **idempotent**. Re-running `-setup` backs up the existing `.env` to `.env.backup` before overwriting. Your data directory and blockchain state are never touched.

4.4 Example .env Output

Auto-generated .env

```

1 # Q-NarwhalKnight Node Configuration
2 # Auto-generated by setup wizard on 2026-03-01 14:30:00 UTC
3
4 # Network
5 Q_NETWORK_ID=mainnet-genesis
6 Q_DB_PATH=./data-mainnet-genesis
7
8 # Node admin wallet (controls admin panel)
9 Q_ADMIN_WALLET=qnka1b2c3d4e5f6...
10
11 # Safety & Sync
12 Q_PREFLIGHT_CHECK=1
13 Q_TURBO_SYNC=1
14 Q_BATCHED_WRITES=1
15 Q_STATE_SYNC=1
16 Q_IS_VALIDATOR=true
17 Q_P2P_PORT=9001
18
19 # Hardware-tuned (auto-detected)
20 ROCKSDB_BLOCK_CACHE_MB=2048
21
22 # Recommended by bootstrap server
23 Q_GOSSIPSUB_HEARTBEAT_MS=300
24 Q_TURBO_CHUNK_SIZE=500

```

4.5 Systemd Service

The generated service file uses `EnvironmentFile=` to load `.env`, making it easy to tune settings without editing the unit file:

Managing the service

```

1 # Start the node
2 sudo systemctl start q-api-server
3
4 # View logs
5 journalctl -u q-api-server -f
6
7 # Edit configuration (then restart)
8 nano .env
9 sudo systemctl restart q-api-server

```

5 Admin Wallet Configuration

The **admin wallet** is the operator's on-chain identity. All operator fee rewards are credited to this address. It is set once at startup and persists in-memory for the lifetime of the process.

5.1 Setting the Admin Wallet

There are two methods, listed in priority order:

1. **CLI flag** (highest priority):
`./q-api-server -admin-wallet abcd1234...`

2. Environment variable:

Q_ADMIN_WALLET=abcd1234...

If neither is provided, the node defaults to the *founder wallet*, meaning all dev fee revenue goes to the project founder. To earn operator rewards, you **must** set your own wallet address.

5.2 Address Format

- 64 hexadecimal characters (32 bytes), case-insensitive.
- Optional qnk or qug prefix is automatically stripped.
- Example: a1b2c3d4e5f6... (64 chars total).

5.3 Verification

After starting the node, verify your admin wallet is active:

Check admin status

```
1 curl -s http://localhost:8080/api/v1/admin/is-admin
2 # Returns: {"is_admin": true, "wallet": "a1b2c3d4..."}
```

6 Fee Structure & Economics

Every block produced by the network includes **coinbase transactions** that distribute the block reward. The fee split is enforced at the consensus level — blocks without correct fee transactions are rejected.

6.1 Fee Flow Diagram

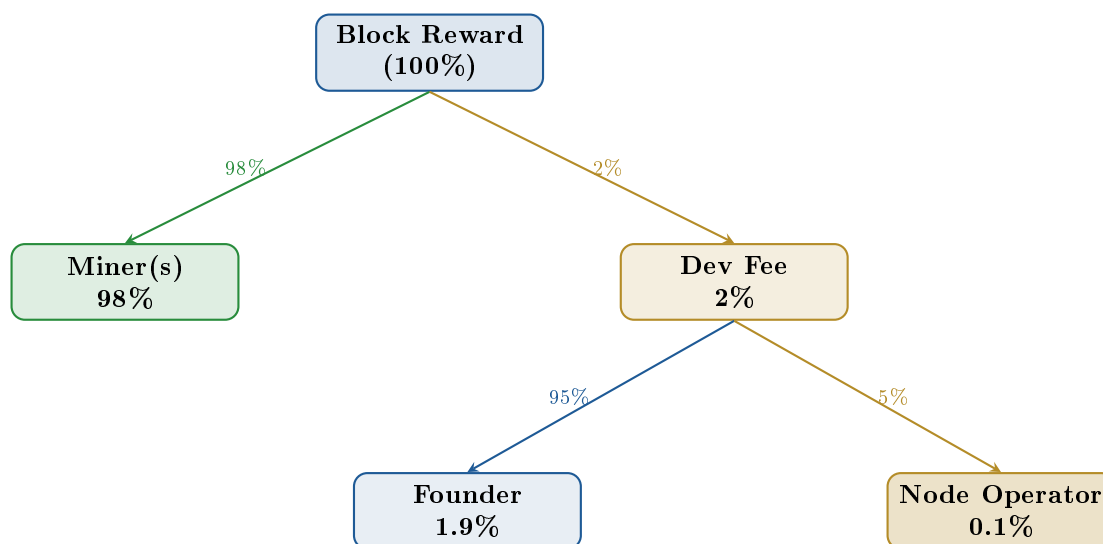


Figure 2: Block reward distribution flow.

6.2 Fee Breakdown

Recipient	% of Reward	BPS	Example (1.0 QUG block)
Miner(s)	98.0%	9800	0.980000 QUG
Founder	1.9%	190	0.019000 QUG
Node Operator	0.1%	10	0.001000 QUG
Total	100.0%	10000	1.000000 QUG

Table 5: Fee breakdown with example amounts.

6.3 How Coinbase Transactions Encode the Split

Each block contains up to three coinbase transactions:

1. **Founder fee TX** – `from=0x00..00`, `to=founder_wallet`, `amount = reward × 200/10000 × 950/1000 = 1.9%` of reward.
2. **Operator fee TX** (if admin wallet configured) – `from=0x00..00`, `to=admin_wallet`, `amount = reward × 200/10000 × 50/1000 = 0.1%` of reward.
3. **Miner reward TX(s)** – one per mining solution, `amount = (reward – dev_fee) / num_solutions`.

If no admin wallet is configured, the entire 2% goes to the founder wallet.

6.4 DEX Protocol Fee

In addition to block rewards, node operators earn from the built-in decentralised exchange:

- **0.05%** (5 basis points) of every swap is collected as a protocol fee.
- This fee is separate from the AMM liquidity-provider fee.
- Revenue accrues to the node that processes the swap.

6.5 Tracking Earnings

Query your cumulative operator earnings via the admin API:

Check operator fee earnings

```
1 curl -s http://localhost:8080/api/v1/admin/fee-earnings
2 # {
3 #   "session_earnings_qug": "0.042000",
4 #   "total_earnings_qug": "1.250000",
5 #   "fee_tx_count": 1250,
6 #   "fee_rate_promille": 50
7 # }
```

7 OAuth2 Device Login — Web Wallet Linking

Miners connect to your node and submit proof-of-work solutions. Before mining begins, the miner must link its identity to a quillon.xyz web wallet via the **OAuth2 Device Authorization Grant** flow (RFC 8628).

7.1 Purpose

- Links the miner's hardware to a verified wallet address.
- Mining rewards are sent to the linked wallet.
- No private keys are exchanged — only the public wallet address.

7.2 Flow Diagram

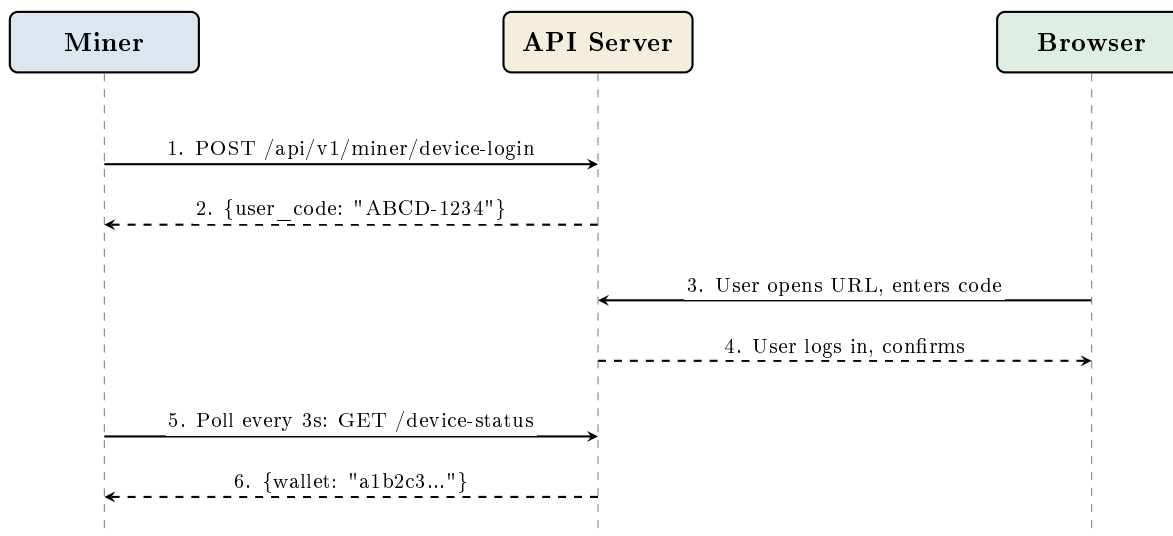


Figure 3: OAuth2 Device Authorization flow for miner wallet linking.

7.3 Step-by-Step

1. Miner requests a device code:

```

1 POST /api/v1/miner/device-login
2 # Response:
3 {
4   "user_code": "ABCD-1234",
5   "verification_url": "https://quillon.xyz/device",
6   "expires_in": 600
7 }
  
```

2. Miner displays the code to the user (TUI or GUI).
3. User opens the URL in a browser and enters the code.
4. User logs in to their quillon.xyz account (creates one if needed).
5. Miner polls every 3 seconds:

```

1 GET /api/v1/miner/device-status?code=ABCD-1234
2 # Pending: {"status": "pending"}
3 # Complete: {"status": "complete", "wallet": "a1b2c3d4..."}
  
```

6. Mining begins to the linked wallet address.

7.4 Security

- Device codes expire after **10 minutes**.
- Wallet address format is validated (64 hex characters).
- The flow never exposes private keys — only the public address is transmitted.
- Rate-limited to prevent brute-force code guessing.

8 Admin API Reference

All admin endpoints are accessible on `http://localhost:8080`. Some require the request to originate from the admin wallet's authenticated session.

Method	Endpoint	Description
GET	<code>/api/v1/admin/is-admin</code>	Check if current session is admin
GET	<code>/api/v1/admin/settings</code>	Retrieve node configuration
GET	<code>/api/v1/admin/fee-earnings</code>	Operator fee earnings (session + total)
GET	<code>/api/v1/admin/operator-fees</code>	Detailed fee history
GET	<code>/api/v1/admin/dev-fee</code>	Current dev fee rate (BPS)
GET	<code>/api/v1/node/info</code>	Node version, height, peers, uptime
GET	<code>/api/v1/node/update-check</code>	Check for binary updates
POST	<code>/api/v1/admin/settings</code>	Update node settings (admin only)

Table 6: Admin API endpoints.

9 Node TUI Dashboard

The QNK node includes a terminal user interface (TUI) for real-time monitoring. Launch with `-tui` or press keys while the node is running in the foreground.

9.1 Key Bindings

Key	Action
D	Dashboard (default view)
L	Live log viewer
N	Network peers and P2P status
S	Statistics (hashrate, blocks, rewards)
F	Physics simulation dashboard
B	Bounty program panel
M	Menu / settings
T	Toggle network throttle mode
Q	Quit gracefully

Table 7: TUI key bindings.

9.2 Operator Earnings Display

When an admin wallet is configured, the TUI header bar shows:

TUI header example

```

1 QNK Node v8.6.2 | Height: 58,432 | Peers: 12
2 Operator: a1b2...c3d4 | Earned: 0.042 QUG (session) | 1.25 QUG (total)

```

The blockchain panel also displays per-block operator fee credits as they occur.

10 Monitoring & Troubleshooting

10.1 Checking Fee Credits

journalctl commands

```

1 # Check operator fee credits
2 journalctl -u q-api-server --since "1_hour_ago" \
3   | grep -E "Operator_fee|OPERATOR_FEE"
4
5 # Check block production
6 journalctl -u q-api-server --since "5_min_ago" \
7   | grep -E "Block.*produced|NEW_BLOCK"
8
9 # Check P2P connectivity
10 journalctl -u q-api-server --since "5_min_ago" \
11   | grep -E "peer.*connected|Gossipsub"

```

10.2 Common Issues

Symptom	Solution
No peers connecting	Ensure port 9001 is open (TCP + UDP). Check firewall and NAT. Verify Q_NETWORK_ID matches the network.
Sync stuck	Check disk space. Restart with ROCKSDB_BLOCK_CACHE_MB=512 if low on RAM. Verify bootstrap peer is reachable.
Missing operator rewards	Confirm -admin-wallet is set and differs from founder wallet. Check /api/v1/admin/is-admin.
OOM killed	Add swap: fallocation -l 4G /swapfile && mkswap /swapfile && swapon /swapfile. Set ROCKSDB_BLOCK_CACHE_MB to 512-2048.

Table 8: Common issues and solutions.

10.3 HA Deployment (Multi-Node Operators)

Operators running multiple nodes can use the included HA deployment script for zero-downtime rolling upgrades:

Rolling upgrade

```

1 # Build new binary
2 cargo build --release --package q-api-server
3
4 # Rolling deploy across all nodes
5 ./scripts/ha-deploy.sh full -y
6
7 # Check status of all nodes
8 ./scripts/ha-deploy.sh status

```

11 Security

11.1 Signing Key Locality

The node's libp2p identity key and any signing keys are generated locally and **never leave the machine**. There is no key upload, no cloud HSM dependency, and no custodial risk. Your

node's identity is sovereign.

11.2 Founder Wallet Protections

The founder wallet (`efca...0723`) has additional consensus-enforced safeguards:

- **Timelock:** Large withdrawals require a multi-block confirmation delay.
- **Vesting schedule:** Founder allocation unlocks gradually over 4 years.
- **Max withdrawal cap:** Single-transaction withdrawal limits prevent rug-pulls.

11.3 Network Isolation

Each network instance (`mainnet2026.1.1`, `testnet`, etc.) is isolated by three mechanisms:

1. **Gossipsub topic namespacing:** `/qnk/{network_id}/blocks`
2. **Protocol handshake validation:** Peers reject mismatched `network_id`
3. **HTTP state sync:** `FullStateSnapshot` includes `network_id` field

This prevents cross-chain contamination even if nodes from different networks share the same physical machine.

References

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